GEOGRAPHIC DATA VISUALIZATION USING TABLEAU

Literacy Rates of Indian States and Union Territories
Census 2011

Project type	Micro Project – Data Exploration and Visualisation
Visualization tool	Tableau Desktop / Tableau Public
Dataset	35 administrative units: 28 states and 7 union territories
Main output	Point map, literacy-rate bar chart, and combined dashboard
Repository suggestion	tableau-literacy-rate-visualization-india

1. Project overview

The project uses geographic visualization to compare Census 2011 literacy-rate percentages for Indian states and union territories. The map provides geographic context; the bar chart provides a clearer comparison of exact values; the dashboard combines both views.

Dataset fields

Field	Meaning / use
State_or_UT	Name of the state or union territory; main geographic field
Administrative_Unit	State or Union Territory; category for grouping
Country	India; geographic context
Literacy_Rate_Percent	Numeric literacy-rate percentage; main measure
Census_Year	Census year (2011)
Source	Dataset source/reference text

The uploaded report describes 35 records, comprising 28 states and 7 union territories. The literacy values range from 61.8% to 94.0%.

2. Prepare the dataset

1. Locate the CSV file used for the project. It should contain the six fields listed above and 35 data rows.
2. Open the CSV in Excel or a text editor and check that the first row contains the column names.
3. Confirm State_or_UT and Administrative_Unit are text fields; Literacy_Rate_Percent is numeric; Census_Year is a whole number.
4. Check for blank cells and duplicate rows. The uploaded report records zero missing cells and zero exact duplicate rows.
5. Keep the country field as India. Do not silently replace the Census 2011 administrative units with newer boundaries.
6. Save a clean copy as `india_literacy_2011.csv` so it is easy to identify when connecting Tableau.

3. Connect the CSV to Tableau

1. Open Tableau Desktop or Tableau Public.
2. On the start page, under Connect, choose Text File and select `india_literacy_2011.csv`.
3. In the Data Source page, inspect the preview and verify that all six column names appear.
4. Check the data types shown by Tableau. If Literacy_Rate_Percent is read as text, change its data type to Number (Decimal).
5. Open a worksheet (for example, Sheet 1). Confirm that the data pane shows State_or_UT and Literacy_Rate_Percent.

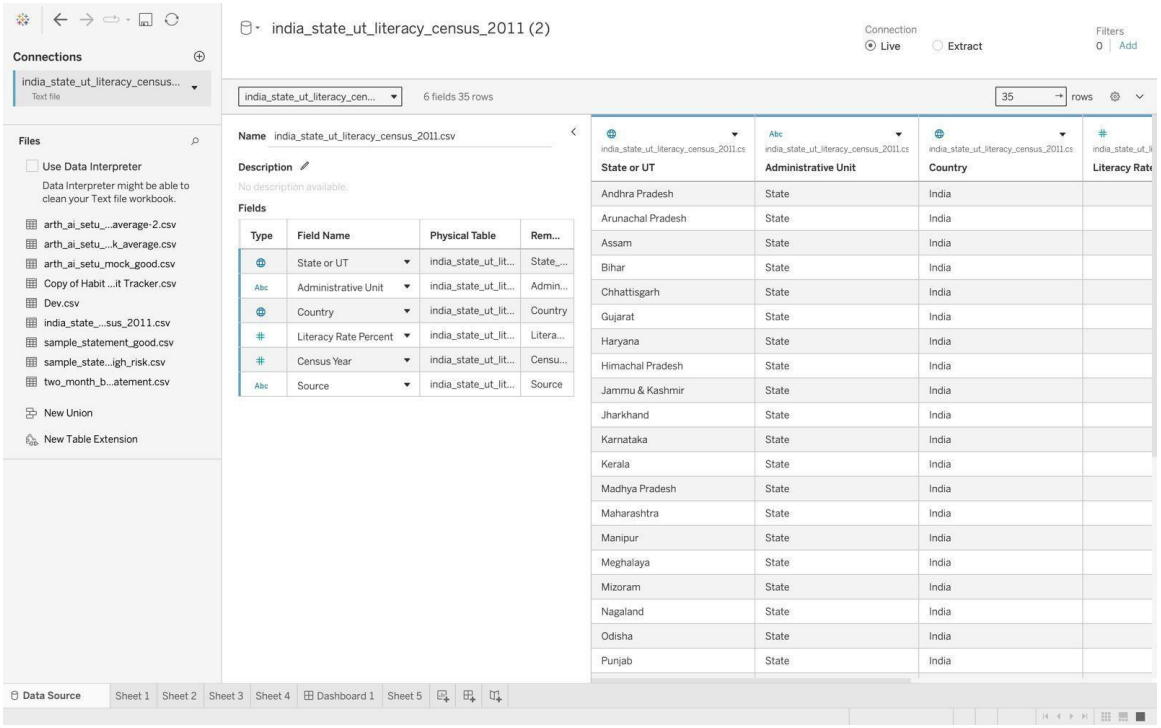


Figure 1. Data-source screenshot extracted from the supplied report (Tableau interface).

4. Create the point map

1. Open a new worksheet and rename it `Literacy Rate Map`.

2. If Tableau does not recognize State_or_UT geographically, right-click the field and select Geographic Role. Choose the most suitable geographic role available. State/Province is generally the first option to test.

3. Use Country as additional context where available. If needed, create or use a Country field containing India for every row.

4. Double-click State_or_UT or drag it into the view to let Tableau generate a map.

5. On the Marks card, use Circle if a point map is desired.

6. Drag Literacy_Rate_Percent to Color. Optionally drag the same measure to Size to make larger values visually distinct.

7. Drag State_or_UT and Literacy_Rate_Percent to Tooltip so the location name and exact rate appear when a mark is selected.

8. Add a clear title such as “Census 2011 Literacy Rate by State/Union Territory”. Check that the color legend is visible and understandable.

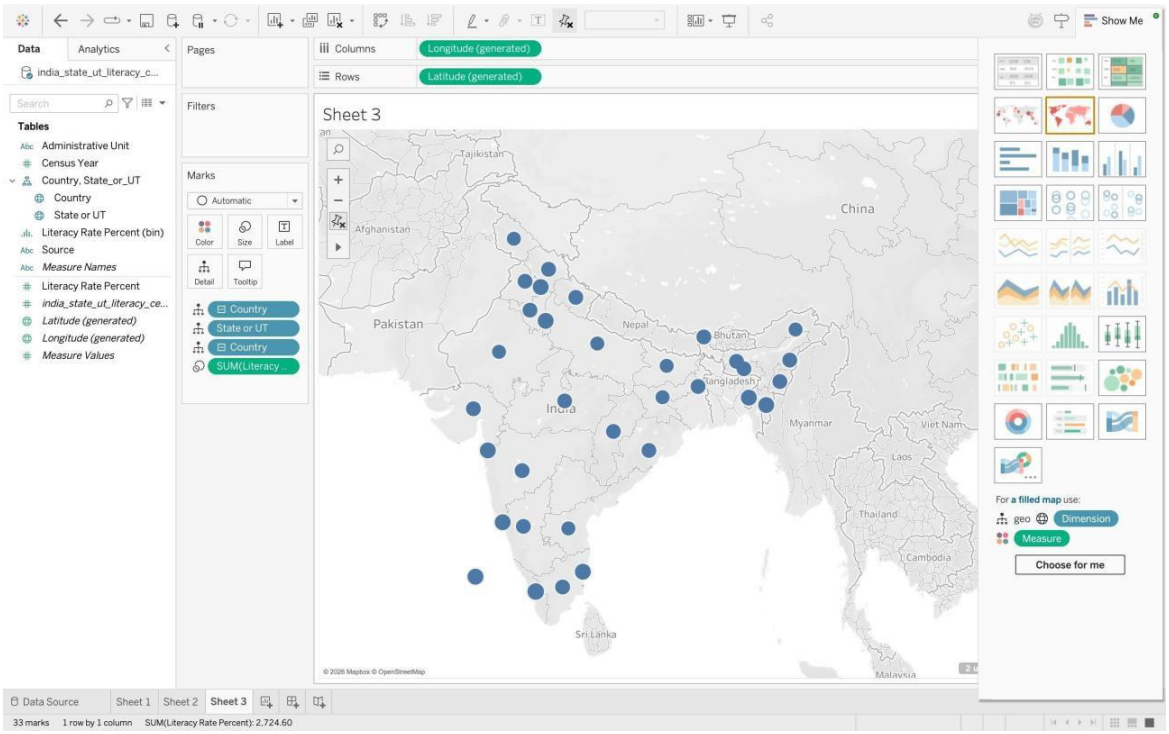


Figure 2. Point-map worksheet shown in the supplied report. The screenshot shows 33 marks and “2 unknown” locations.

5. Resolve unknown locations

1. Click the “unknown” indicator on the map to inspect the unmatched location names.

2. Check the spellings and historical labels against the 2011 dataset. The uploaded report mentions abbreviations such as “A & N Islands” and “D & N Haveli” as possible causes, but it does not prove which two names failed.

3. Try the full official location name in the data, or edit aliases only when you have verified the correct match.

4. If geographic name matching remains unreliable, use a verified latitude/longitude file and assign those fields to generated Latitude and Longitude in Tableau.

5. Confirm the number of plotted marks after correction. If two records remain unknown, mention that limitation in the report rather than hiding it.

6. Create the literacy-rate bar chart

1. Create a new worksheet and rename it `Literacy Rate by State UT`.

2. Drag State_or_UT to Rows.

3. Drag Literacy_Rate_Percent to Columns. Check that Tableau uses AVG or the appropriate single rate per unit, not SUM of repeated records.

4. Choose Bar from Show Me, or select Bar in the Marks card.

5. Sort the bars in descending order to make high and low values easy to compare.

6. Drag Literacy_Rate_Percent to Label and turn on Show Mark Labels.

7. Format the axis as a percentage or append a clear “Literacy rate (%)” axis title. Use a readable chart title.

8. Check the chart includes all 35 records and that labels are readable. A horizontal bar chart is recommended when state names are long.

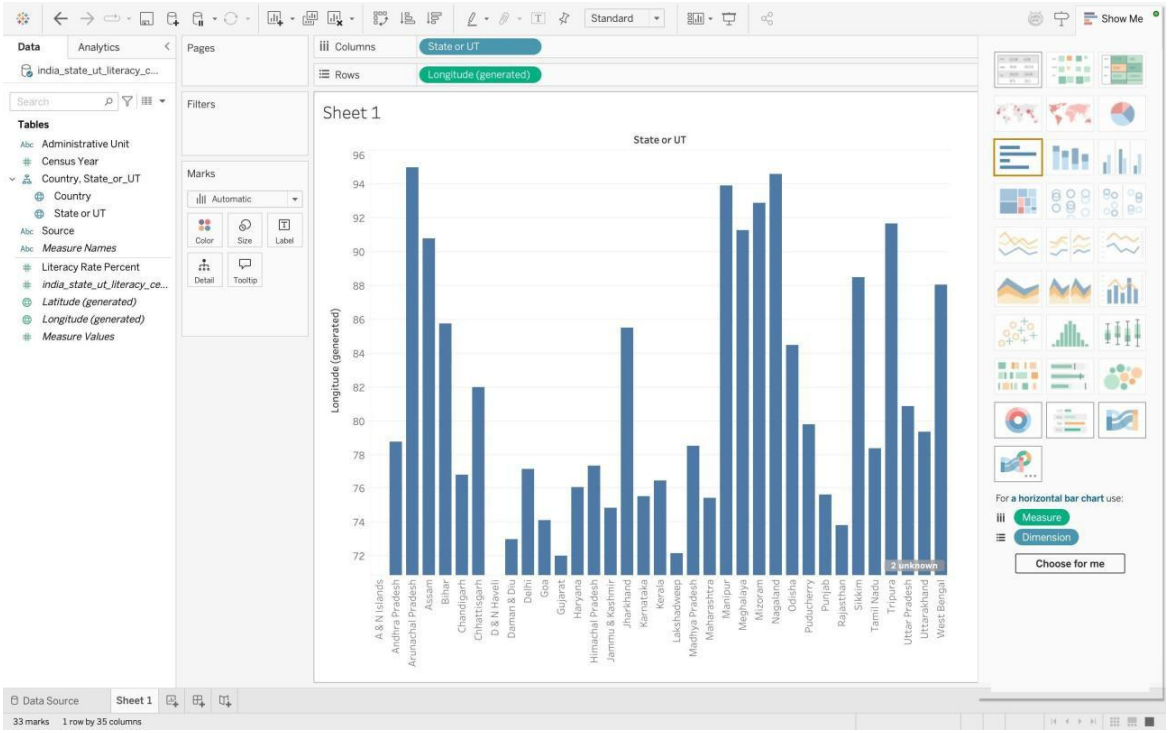


Figure 3. Bar-chart worksheet screenshot extracted from the supplied report.

7. Build the dashboard

1. Click the New Dashboard icon at the bottom of Tableau.

2. Rename the dashboard `India Literacy Dashboard – Census 2011`.

3. Drag the `Literacy Rate Map` worksheet onto the dashboard.

4. Drag the `Literacy Rate by State UT` worksheet beside or below the map.

5. Place the color legend where it does not take too much space. Adjust the map and chart sizes so state names and map marks are readable.

6. Use the dashboard title to state the topic and census year.

7. If useful, open the worksheet menu on the map or chart and select “Use as Filter” so selecting a mark filters the other view.

8. Test the dashboard by selecting a state or mark, clearing the selection, and checking that the views behave as expected.

A good dashboard uses the map for location context and the sorted bar chart for exact comparisons. Avoid a narrow map, an oversized legend, or unreadable state labels.

8. Validate and interpret the output

Measure	Value in uploaded report
Number of administrative units	35
Mean literacy rate	77.85%
Median literacy rate	76.9%
Lowest recorded rate	Bihar — 61.8%
Highest recorded rate	Kerala — 94.0%
Range	32.2 percentage points
States / Union Territories	28 / 7

These are the descriptive values reported in the uploaded PDF. They describe the 2011 records and are not population-weighted national estimates.

• Check the bar chart includes 35 units.

• Check the highest and lowest labels against the dataset.

• Check the map for unknown locations and do not overstate its coverage.

• Check the title, legend, tooltips, and axis labels are readable.

• Take screenshots only after the final output has been checked.

9. Save and export your Tableau work

1. Save the workbook using File → Save As.

2. For a portable workbook that includes local data, choose Tableau Packaged Workbook (`.twbx`) if the option is available.

3. Save a screenshot of the completed dashboard as `tableau_dashboard.png`.

4. Export the dashboard or workbook to PDF if your Tableau version offers a suitable export option. Review the PDF to ensure labels are not clipped.

5. Keep the source CSV and final report alongside the workbook. Do not include confidential data or credentials.

10. Create a GitHub repository and upload the project

1. Open https://github.com/ and sign up or sign in.

2. Click the plus (+) menu and choose New repository.

3. Use the repository name `tableau-literacy-rate-visualization-india`.

4. Enter the description: “Geographic visualization of Census 2011 literacy rates for Indian states and union territories using Tableau.”

5. Choose Public only if you want the project to be publicly visible. Add a README file and create the repository.

6. Open the repository, select Add file → Upload files, and upload the files you actually created.

Suggested repository layout

```
tableau-literacy-rate-visualization-india/
■■■ README.md
■■■ data/
```

```
■ ■■■ india_literacy_2011.csv
■■■ screenshots/
■ ■■■ tableau_dashboard.png
■■■ workbook/
■ ■■■ tableau_project.twbx
■■■ report/
■■■ project_report.pdf
```

7. Write a commit message such as “Add Tableau literacy visualization project” and click Commit changes.

8. Open the repository page and copy its URL. Add that URL to your project report only after the repository has been created.

Do not claim a GitHub repository has been created until you have actually uploaded the files. Replace the example file names with your real file names.

11. Publish an interactive dashboard

1. If you want others to interact with the dashboard online, use Tableau Public at https://public.tableau.com/.

2. Sign in and publish the workbook using the available Tableau Public publishing option.

3. Open the published dashboard and copy its public URL.

4. Add the URL to the GitHub README. Confirm the published workbook contains no private data.

GitHub stores project files; Tableau Public hosts an interactive visualization. They serve different purposes.

12. Final submission checklist

• CSV dataset contains the expected fields and 35 rows.

• Tableau data-source view opens correctly.

• Map and bar chart worksheets are saved.

• Dashboard includes both map and bar chart.

• Unknown locations are resolved or documented.

• Workbook is saved as .twbx when portability is required.

• Dashboard screenshot and final project report are included.

• GitHub repository URL is real and accessible to the intended reviewer.

• All claims and screenshots correctly identify Tableau rather than Power BI.